HW - 2

Part A

1. $\text{Output} = \frac{8 + 2\times0 - 3}{1} + 1 = 6\times6\times4$

2. $\text{Output Shape} = \frac{10 + 2\times1 - 3}{1} + 1 = 10\times10\times2$

3. Parameters per filter = filter size × input channel + bias
$= 3\times3\times1 + 1 = 10$ parameters
Total parameters = Parameters per filter × No. of filters $= 10\times3$
$= 30$ total trainable parameters.

4. $\text{Output Shape} = \frac{3 + 2\times0 - 2}{1} + 1 = 2\times2 \rightarrow$ Dimensions of our output feature map $\rightarrow \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$ through element wise multiplication.

5. $\text{Dimension of output feature map} = \frac{4 + 2\times0 - 2}{2} + 1 = 2\times2$

$\begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix} \rightarrow$ Max values in each 2×2 window over the inp. with a stride of 2.

Part B

6. Filters in a CNN detect specific local features or patterns in the input data. In early layers filters detect simple low-level features like edges, corners, textures, while filters in deeper layers combine these to detect complex high-level features like shapes or specific parts of objects

7. Padding adds extra pixels around the outer border of the input feature map before applying a convolution. It is done to control the spatial dimensions of the output, preventing the feature map from shrinking too fast, and to ensure the features at the edges of the input image are considered equally and not washed away during the actual convolution process.

8. Maxpooling downsamples the input by reducing the spatial dimensions while retaining the most prominent features. This reduces computational load, reduces the number of parameters to prevent overfitting, and provides a degree of spatial translation invariance, which means the network can still recognize a feature even if it shifts slightly in the input.